

1 description

still trying to express f_s and f_e in a better way

Let p denote fraction of infected patches and N denote population density of infected patches.

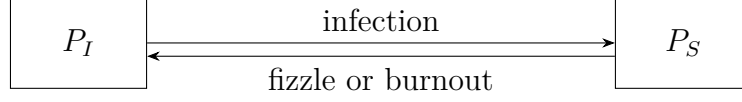


Figure 1: patch state transition

spread: Hosts move around and simultaneously bring pathogens to susceptible patches (combining movement and infection in one step to avoid calculating average density again)

$$f_s : \binom{p}{N} \rightarrow \binom{p + Bp(1-p)}{N + c(1-p)(S-N)}$$

B is infection rate (currently setting B as constant since we've already set R_0 as constant too), c is moving probability of individual rats, and S is the population density of susceptible patches (assumed constant).

epidemic: Infected hosts die and so some infected patches burnout.

$$f_e : \binom{p}{N} \rightarrow \binom{P_1 p}{(1-zD)N}$$

$P_1 = P1_prob(N)$ is the persistence probability, z is final size of this epidemic wave, D is the death probability of infected hosts.

growth: Assume a constant birth rate r .

$$f_g : \binom{p}{N} \rightarrow \binom{p}{(1+r)N}$$

The overall iteration process is

$$f = f_g \circ f_e \circ f_s : \binom{p}{N} \rightarrow \binom{P_1(p + Bp(1-p))}{(1+r)(1-zD)(N + c(1-p)(S-N))}$$

where $P_1 = P1_prob(N + c(1-p)(S-N))$

2 range of parameters

Consider quadratic function $y = x + kx(1-x)$ where $0 \leq x \leq 1$ and $k \geq 0$,

$$f(x) = x + kx(1-x) = -kx^2 + (k+1)x, x \in R$$

We want to make sure that $f([0, 1]) \subset [0, 1]$. The function attains its maximum at $x^* = \frac{k+1}{2k}$. Note that $f(0) = 0$ and $f(1) = 1$, so we'll need to require that $x^* \geq 1$, i.e. $0 \leq k \leq 1$.

- **infection:** $p \leftarrow p + Bp(1 - p)$, we require $0 \leq B \leq 1$
- **burnout:** $p = P_1 p$, $0 \leq P_1 \leq 1$
- **epidemic:** $0 \leq z \leq 1$ and $0 \leq D \leq 1$
- **movement:** $0 \leq c \leq 1$
- **birth:** $(1 + r)(1 - zD) \leq 1$

3 Disease Free Equilibrium

If we start at a point where $p > 0, N > 0$, all variables remain positive during the iterations as each step maps positive values to a positive values.

If we start at $p = 0, N > 0$ then population size ends up at

$$N_t = N_{t+1} = (1 + r)(1 - zD)(N_t + (p - 1)(S - N_t))$$

i.e.

$$N_{DFE} = \frac{c(1 + r)(1 - zD)S}{1 - (1 + r)(1 - zD)(1 - c)}$$

N_{DFE} increases with r, c, S and decreases with z, D .

4 invasion analysis

$$J_{f_s} = \begin{pmatrix} 1 + B - 2Bp & 0 \\ -c(S - N) & 1 \end{pmatrix}$$

$$J_{f_e} = \begin{pmatrix} P_1 & p \frac{\partial P_1}{\partial N} \\ 0 & 1 - zD \end{pmatrix}$$

$$J_{f_g} = \begin{pmatrix} 1 & 0 \\ 0 & 1 + r \end{pmatrix}$$

And at DFE we have $p = 0, N = N_{DFE}$ so the matrices above are all lower triangular, so eigenvalues of $J_f = J_{f_g} J_{f_s} J_{f_e}$ are $e_1 = (1 + B)P_1$ and $e_2 = (1 - zD)(1 + r)$. We required previously that e_2 is always smaller than 1. If $(1 + B)P_1|_{N=N_{DFE}+c(S-N_{DFE})} < 1$, DFE is stable, so the disease cannot persist; if it is greater than 1, DFE is unstable, the disease will persist and system ends up at endemic equilibrium.

5 Endemic equilibrium

Endemic equilibrium indicates

$$\begin{aligned} P_1(1 + B(1 - p)) &= 1 \\ (1 + r)(1 - zD)(N + c(1 - p)(S - N)) &= N \end{aligned}$$

$$P_1 = p_k(1 - q(x_{in})^{y_\star(N+c(1-p)(S-N))})$$

$p_k, q(x_{in}), y_\star$ are all defined in burnout paper. Denote $\frac{1}{(1+r)(1-zD)} = t$ then we get $N + c(1 - p)(S - N) = tN$ and $1 - p = \frac{N(t-1)}{c(S-N)}$. Plug this into first equation and we get

$$(1 - e^{ty_\star \ln q(x_{in})N})(1 + \frac{BN(t-1)}{c(S-N)}) = \frac{1}{p_k}$$

i.e.,

$$e^{ty_\star \ln q(x_{in})N} \frac{N - \frac{cS}{c-B(t-1)}}{N - \frac{cS(p_k+1)}{c(p_k+1)-p_k B(t-1)}} = \frac{p_k B(t-1) - c(p_k+1)}{p_k(B(t-1) - c)}$$

denote $w = ty_\star \ln q(x_{in}), u = \frac{cS}{c-B(t-1)}, v = \frac{cS(p_k+1)}{c(p_k+1)-p_k B(t-1)}, a = \frac{p_k B(t-1) - c(p_k+1)}{p_k(B(t-1) - c)}$ Then the equation is

$$e^{wN} \frac{N - u}{N - v} = a$$

The solution is

$$N^* = u + \frac{1}{w} W_{-ae^{-wu}}(wae^{-wu}(u - v))$$

$$p^* = 1 - \frac{N^*(t-1)}{c(S - N^*)}$$

W_r is r-Lambert function, it is the inverse of the function $xe^x + rx$.